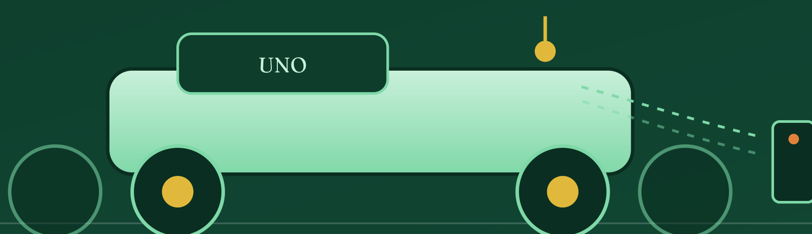
ROBOTICS & EMBEDDED SYSTEMS

PROJECT REPORT

Autonomous Electronics Project

IR Remote-Controlled 4WD Smart Robot Car

A four-wheel-drive robotic platform built on an Arduino UNO, driven by an L298N dual H-bridge motor controller and steered wirelessly through infrared commands from a handheld remote.



PROJECT TEAM

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Platform — Arduino UNO
Control — Infrared Remote
Drive — 4WD Differential Steering

Contents

Report Overview

A complete walk-through of the build — from concept and objectives to hardware, wiring, control logic and results.

> *"A small platform of four motors and one microcontroller — enough to turn a line of code into motion across a room."*

01 · Introduction

About the Project

What the robot is, why it was built, and how it fits together.

The **IR Remote-Controlled 4WD Smart Robot Car** is a compact mobile robotics platform designed to demonstrate the core building blocks of embedded control: a microcontroller brain, a motor driver muscle, and a wireless sensory input. The chassis carries four independently mounted DC gear motors on a laser-cut MDF frame, giving the robot strong torque and stable four-wheel traction on indoor and outdoor floors alike.

At the centre of the build sits an **Arduino UNO**, which reads incoming infrared signals from a handheld 21-button remote and translates each keypress into a driving command. Because the UNO cannot supply enough current to run four motors directly, an **L298N dual H-bridge motor driver module** sits between the microcontroller and the motors, amplifying the UNO's low-power logic signals into the higher current needed to spin the wheels in either direction.

Power for the entire system comes from a pair of rechargeable **3.7V 18650 lithium-ion cells** wired in series inside a battery holder mounted on the chassis deck, making the robot fully self-contained and untethered. An **infrared receiver module (HW-477)** constantly listens for signals from the remote, decodes the button pressed, and passes that value to the Arduino over a single signal wire — no Bluetooth, no Wi-Fi, just line-of-sight infrared, the same technology used by a television remote.

Drive System

Four DC gear motors, driven in pairs (left side / right side) for differential steering — forward, reverse, left turn, right turn and stop.

Control Method

A 21-key infrared remote sends unique hex codes for each button, captured wirelessly by the onboard IR receiver.

Power Supply

Two 18650 Li-ion cells (3.7V each) in series, feeding both the motor driver and the Arduino's regulated 5V rail.

Chassis

A two-layer laser-cut 4WD chassis kit that houses the electronics between its plates and mounts all four motors.

02 · Objectives

Project Goals

What we set out to achieve and why each goal mattered to the final build.

1 **Build a reliable four-wheel-drive base**

Assemble a sturdy chassis capable of carrying the electronics safely while providing enough torque and grip to move confidently across different surfaces.

2 **Achieve smooth wireless driving control**

Use infrared remote control instead of physical buttons or a tethered connection, so the robot can be steered from a distance, line-of-sight, without extra hardware complexity.

3 **Correctly interface a motor driver with the microcontroller**

Learn and apply H-bridge motor driving so the Arduino's low-current digital pins can safely control high-current DC motors in both directions.

4 **Make the robot fully self-powered and portable**

Run the entire electronics stack from onboard rechargeable batteries, with no external power cable, so the robot can be tested anywhere.

5 **Write clean, extendable firmware**

Structure the Arduino code around clear motor functions (forward, reverse, left, right, stop) and a simple IR-command lookup, so new behaviours can be added later.

6 **Understand the full stack of a mobile robot**

Gain hands-on experience spanning mechanical assembly, power electronics, digital control and embedded software — the four pillars of any robotics project.

03 · Hardware

Components & Hardware

Every part that went into the build, and the role it plays in the system.

Component	Qty	Role in the system
Arduino UNO	1	Main microcontroller — reads IR signals and drives the motor control pins.
L298N Motor Driver Module	1	Dual H-bridge that amplifies Arduino signals to safely power all four motors in either direction.
IR Receiver Module (HW-477)	1	Detects the infrared beam from the remote and outputs a demodulated digital signal to the Arduino.
Infrared Remote Control (21-key)	1	Handheld transmitter — arrow keys, OK, and numeric keys each send a unique code.
DC Gear Motors	4	Drive the wheels; paired left/right for differential steering.
4WD Acrylic / MDF Chassis Kit	1	Two-tier body that mounts the motors, wheels and electronics.
18650 Li-ion Battery (3.7V)	2	Rechargeable power cells wired in series for a ~7.4V supply.
Battery Holder	1	Houses both cells and routes power to the motor driver and Arduino.
Jumper Wires	Several	Signal and power connections between all modules.

Why the L298N?

DC motors draw far more current than an Arduino pin can safely provide, and they need current to flow in either direction to reverse. The L298N contains two H-bridge circuits that solve both problems at once — each pair of digital inputs (IN1/IN2 and IN3/IN4) sets a motor's direction, while the module's own power stage supplies the current, keeping the Arduino electrically isolated and safe.

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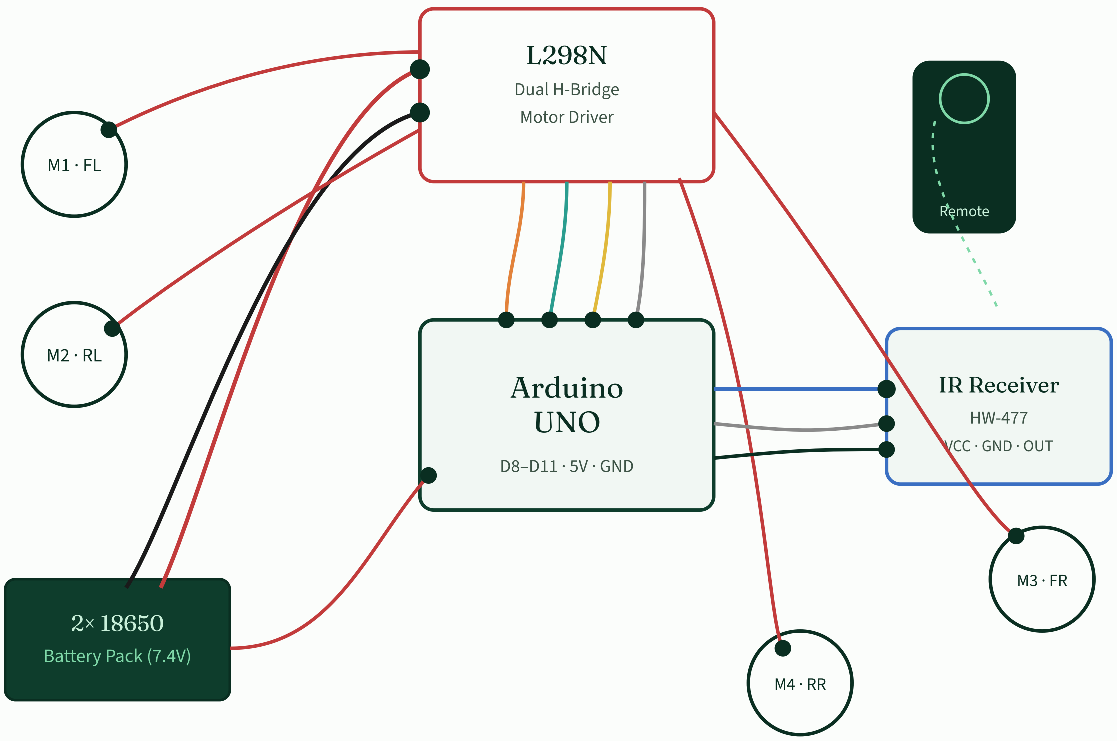
04 · Circuit Design

Wiring Diagram

How power and signal wires connect the Arduino, motor driver, IR receiver, battery and motors.



Schematic block diagram — colours indicate signal type: red/black = power, orange/teal/yellow/grey = motor control logic, blue/grey/black = IR receiver signal lines.

Power Path

Battery pack → L298N (12V screw terminal) and Arduino 5V pin. The L298N's onboard regulator can also feed the Arduino if the jumper is left in place.

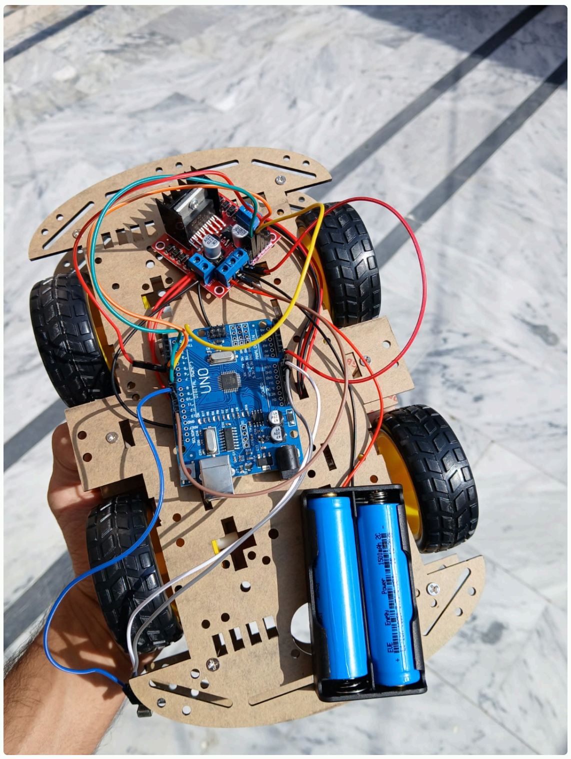
Control Path

Arduino digital pins D8–D11 → L298N IN1–IN4, setting motor direction; the driver's OUT terminals connect straight to the four motors.

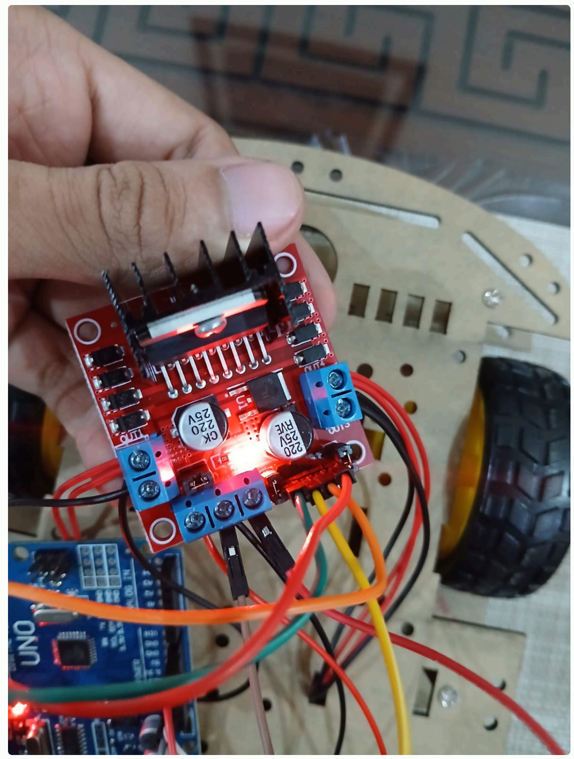
05 · Build Gallery

The Physical Build

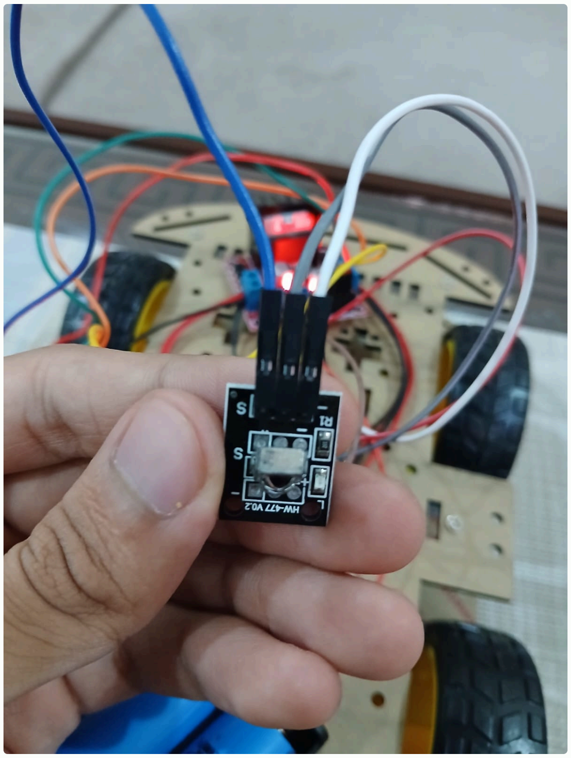
Photos taken during assembly and wiring of the actual robot.



Underside view — Arduino UNO, L298N driver and battery pack mounted on the chassis deck.



L298N dual H-bridge module, powered and connected to all motor lines.



HW-477 infrared receiver module — three-pin connection to the Arduino.

The 21-key infrared remote used to send driving commands.

07

— 06 · Software

Firmware & Control Logic

How the Arduino sketch turns a remote button press into wheel motion. Written in the Arduino IDE, without reproducing the raw code line by line.

The firmware is built around the open-source **IRremote** library, which listens on a dedicated digital pin connected to the IR receiver's output. Every time a button on the remote is pressed, the receiver demodulates the infrared pulses and the library decodes them into a unique hexadecimal command code — for example, the left-arrow key, right-arrow key and OK button each produce their own distinct value.

1. **Setup:** The four motor-control pins (IN1–IN4) are configured as outputs, the motors are commanded to a stopped state, and the IR receiver is initialised to begin listening for signals.
2. **Listen:** The main loop continuously checks whether a new infrared command has arrived from the remote.
3. **Decode:** When a signal is received, its hexadecimal code is compared against a set of named constants — one for each direction the robot understands (forward, reverse, left, right, stop).
4. **Match & act:** Once the code matches a known command, the corresponding motor function is called, which sets the correct combination of HIGH/LOW states on IN1–IN4 to spin the motors the right way.
5. **Repeat or stop:** If no new signal arrives, the robot continues its last action; releasing the button (or pressing the centre "OK" key) triggers the stop function, cutting power to all motors.

Motor Direction Logic

Each motor pair is driven by setting one control pin HIGH and its partner LOW to spin forward, and reversing that pattern to spin backward. Turning is achieved by driving the left-side and right-side motor pairs in opposite directions or at different states — for example, the left pair reversing while the right pair moves forward produces an in-place turn.

Library Used

`IRremote.h` handles capturing and decoding the infrared protocol from the remote automatically.

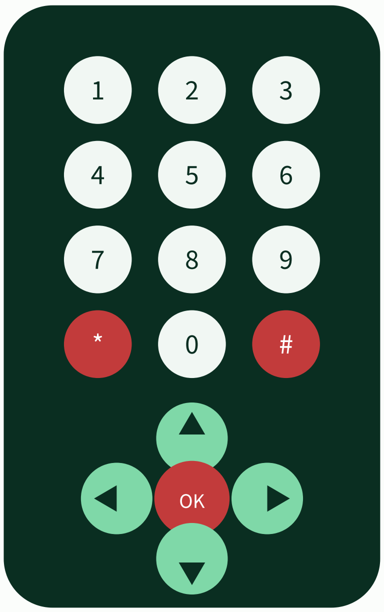
Design Style

Named functions — *moveForward(), moveBackward(), turnLeft(), turnRight(), stopMotors()* — keep the logic readable and easy to extend.

07 · Remote Control

IR Remote Command Map

Each key on the 21-button remote is decoded to a unique hex code and mapped to a driving action.



Key	Action
▲ Up arrow	Drive forward
▼ Down arrow	Drive in reverse
◀ Left arrow	Turn left
▶ Right arrow	Turn right
OK (centre)	Stop all motors
0 – 9, *, #	Reserved for future features

Each key transmits its own unique hexadecimal code; the sketch stores the codes for the arrows and OK button as named constants, so the same physical remote can be reprogrammed for new behaviour just by editing those definitions — without touching the decoding logic.

08 · Verification

Testing & Results

How the robot was validated after assembly, and what we observed.

Test	Expected outcome	Result
Power-on self-check	Arduino boots, driver LED lights, "Robot Ready" message on serial monitor	Pass
Forward command	All four wheels spin forward in sync	Pass
Reverse command	All four wheels reverse together	Pass
Left / right turn	Opposite-side wheel pairs rotate differently to pivot the chassis	Pass
Stop command	All motors cut power immediately	Pass
IR range check	Reliable reception within typical remote line-of-sight range	Pass
Battery run-time	Continuous operation on a single charge of the 18650 pair	Pass

Debugging relied heavily on the Arduino IDE's serial monitor, printing the received hex code for every button press. This made it straightforward to confirm that each key was being decoded correctly before wiring it to a motor action, and to catch cases where stray reflections briefly triggered false readings.

09 · Reflection

Challenges & Future Scope

What slowed us down, and where this platform could go next.

Wiring density

With four motors, a driver board, an IR module and a battery pack sharing a small chassis, keeping wires organised and avoiding shorts required careful routing and labelling.

Power stability

Running motors and logic from the same battery pack caused brief voltage dips under load, addressed by keeping the driver's power input separate from the Arduino's logic supply.

IR signal reliability

Ambient light and indirect angles occasionally caused missed or duplicate readings, mitigated with simple debounce logic in software.

Calibration of turns

Small differences between motor units meant the robot did not always turn symmetrically — fine-tuned through testing.

What's next

- Add ultrasonic obstacle detection for basic autonomous collision avoidance.
- Introduce PWM-based speed control for smoother acceleration and variable-speed turning.
- Upgrade to Bluetooth or Wi-Fi control alongside IR for longer range and app-based driving.
- Add a rechargeable power indicator and low-battery cutoff for safer operation.

10 · Conclusion

A working platform, built from first principles

This project brought together mechanical assembly, motor driving, power management and embedded firmware into a single, self-contained robot. Every subsystem — chassis, driver, receiver, battery and code — was chosen, wired and tested by hand, resulting in a 4WD robot that responds reliably to a simple infrared remote. Beyond the working robot, the build served as a hands-on foundation in the principles that underpin nearly every mobile robotics project: sensing, decision-making and actuation.

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